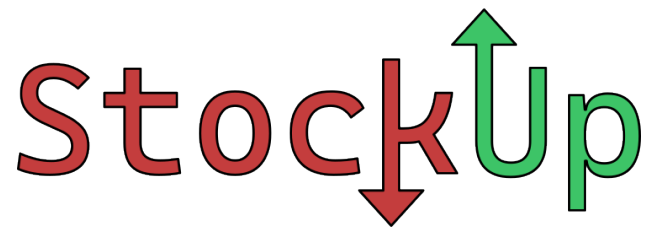


CS 3600 Project Phase 2

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1 Project

StockUp is a social and financial app which is a combination of fantasy football and real-world investment. On StockUp, users will be able to join a team, invest in stocks, and earn money for their team.

2 System Design

2.1 Database

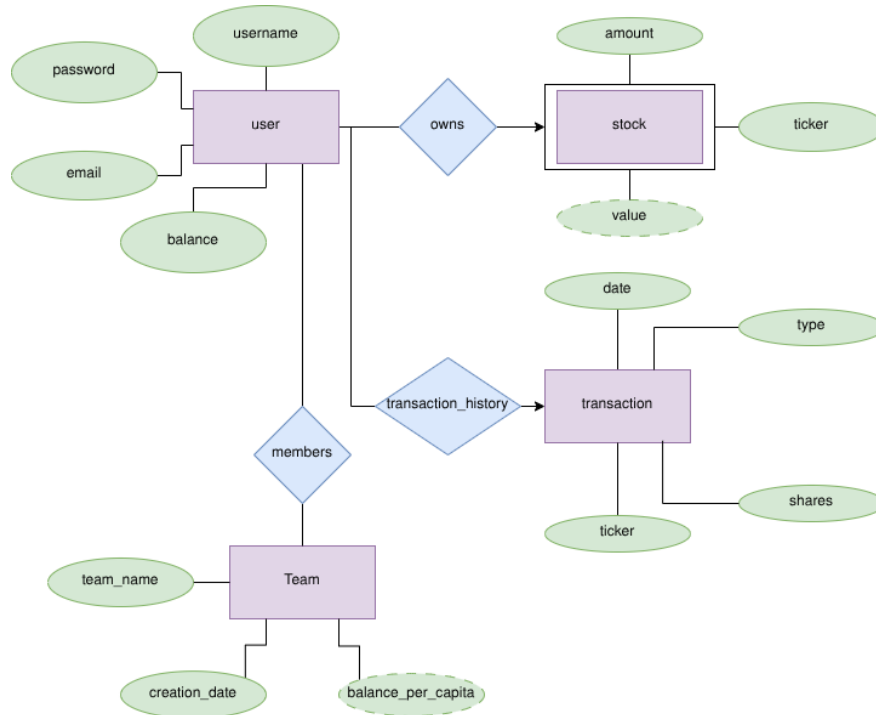


Figure 1: ER Diagram

Django and SQLite3 are used to create and query the database. The tables are generated by Django with the models.py file. Any changes made to the database model are applied by executing the migrate command with manage.py.

2.2 Backend

The backend routes are routed by Django. All routes are implemented in the views.py file. The routing URLs are assigned in the urls.py file from implementations in the views.py file.

2.3 Frontend

The frontend is designed with HTML and Bootstrap templates generated with Django. Templates are stored in the templates directory with static resources from the static directory. Essentially, HTML files are stored in the templates directory and CSS/JS files are stored in the static directory for sharing.

2.4 Stock Predictor

The stock information is gained from the Yahoo Finance package, `yfinance`. The stock predictor model is implemented with `XGBoost` and stored as a file that is restored with the `joblib` package. The `joblib` package allows for the model to be trained once to reduce prediction times.

A prediction may be obtained through a POST request sent to the `invest` route. The POST request body requires a ticker symbol and a sector.

3 Functional Components

3.1 Database

The database is fully implemented.

3.2 Backend

The core features of the backend are implemented with routes, database, and pages.

3.3 Frontend

User interface is complete and the content is implemented. The current interface is clean and modern. However, additional refining is desired for improved ergonomics.

3.4 User Login

The user login and sign up forms allow a user to sign into their account or create a new account with their email address and desired password.

3.5 Stock Predictor

The stock predictor is partially implemented. The user may search a particular stock, and select the stock industry to view the closing price history of the stock over the course of the past two-hundred days. Once the information is submitted, a graph appears that shows the stock price history along with the 30 day price prediction to help the user to gauge whether the stock is worth purchasing. After a decision is made by the user and the `invest` option is selected, the user is charged and the stock becomes owned by the user's `StockUp` account.

3.6 Leaderboards

The leaderboards show the top ranking teams based on the value per capita. In the team view, users may fill a search bar text box and submit the search result to view teams that match the specified search. Each element in the list of teams shows the team's name, number of members, and the balance per capita. The user may select a team to view its details.

3.7 Team View

When viewing the details of a team, the user is presented with the team's number of members, balance per capita, and total balance. Additionally, a graph shows each member of the team displayed as a node with name that is connected to the team node. The user may request to join the team by selecting the `join team` option.

3.8 Administration Utilities

There are administration tools that allow the server administrators to view and modify database values and tables. The utilities are protected with an admin login that requires user privileges to access. These utilities allow for easy debugging and error tracking to resolve any potential issues.

4 Validation

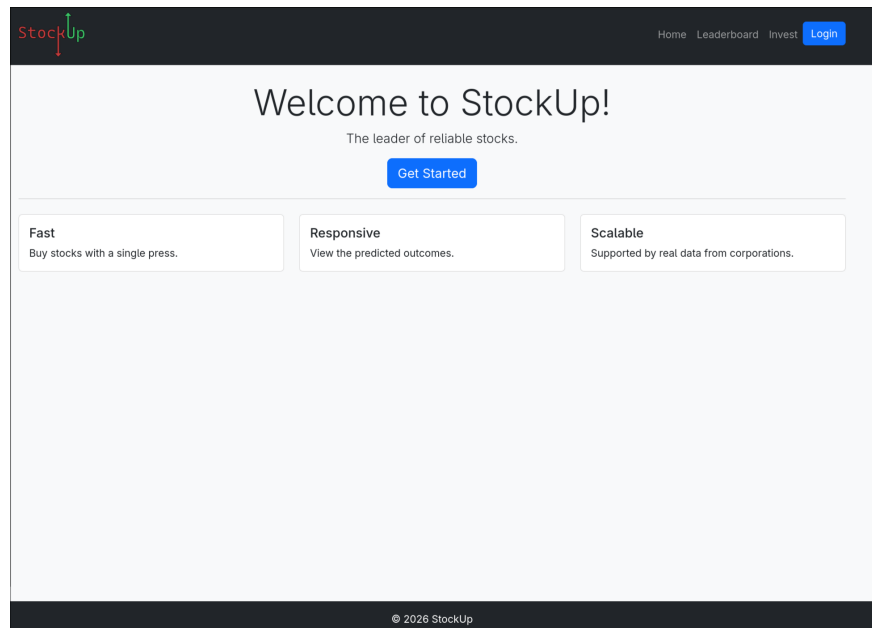


Figure 2: Home Page

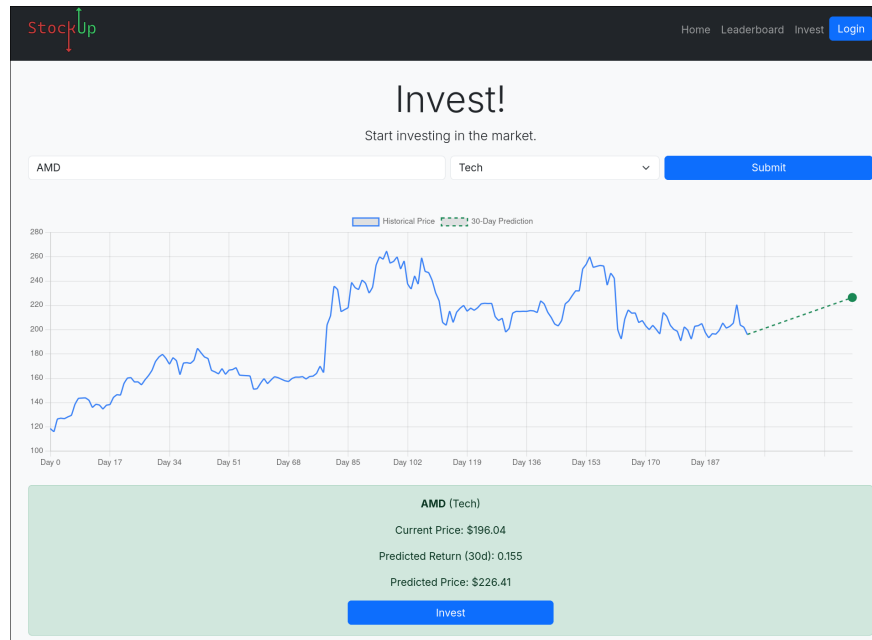


Figure 3: Stock Investing Page

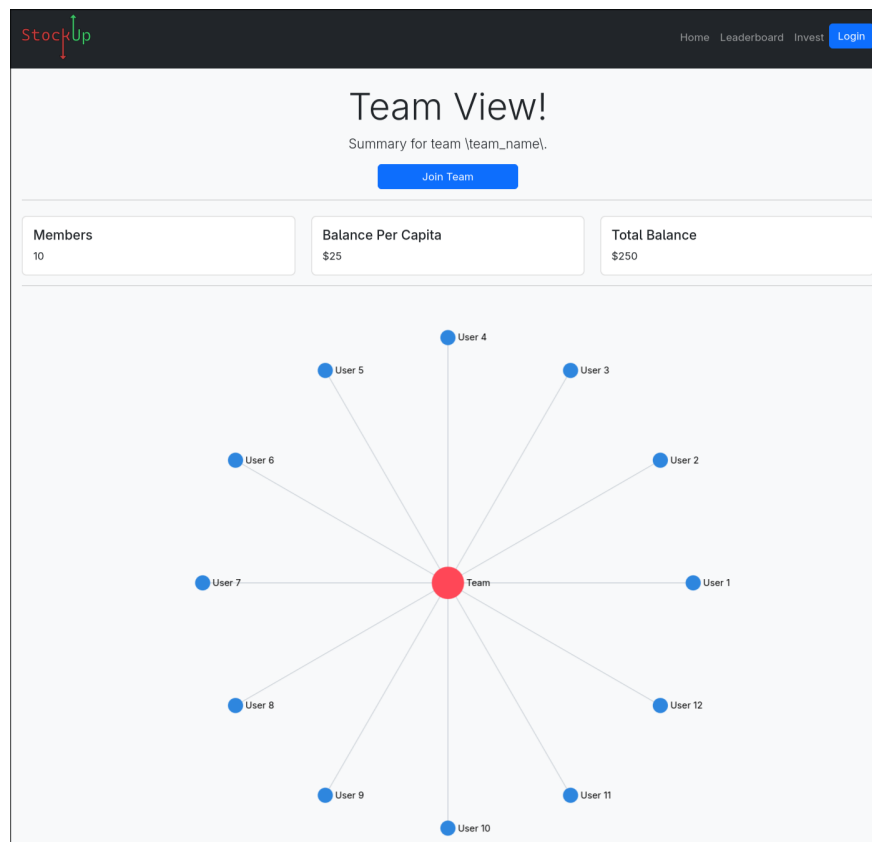
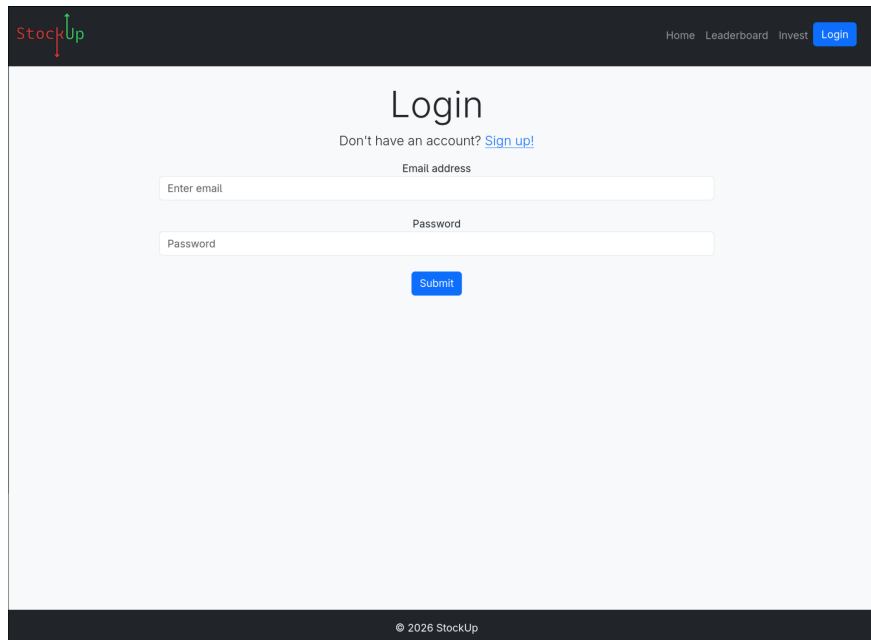
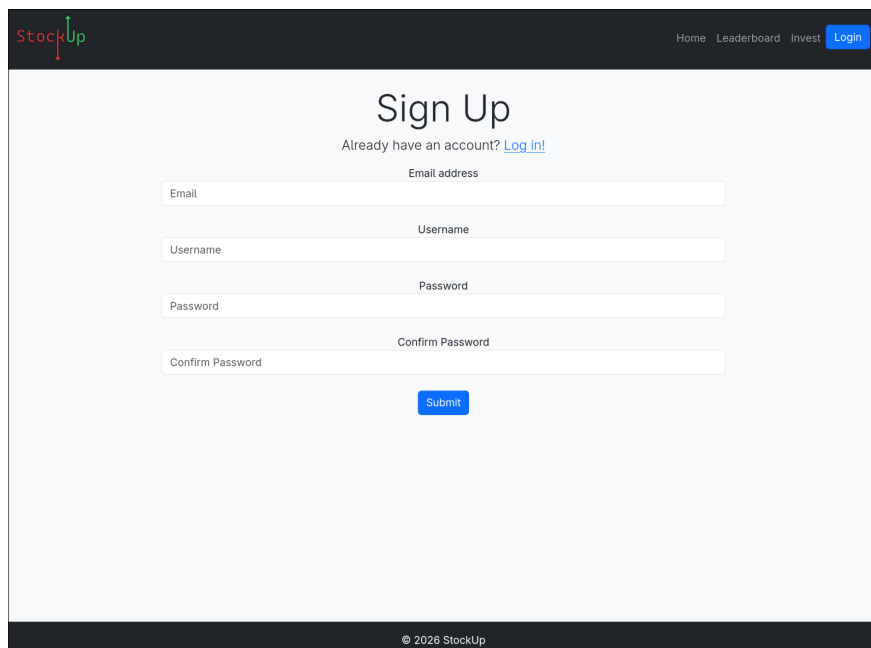


Figure 4: Team View Page



The screenshot shows the 'Login' page of the 'StockUp' application. The header is dark with the 'StockUp' logo on the left and navigation links 'Home', 'Leaderboard', 'Invest', and a 'Login' button on the right. The main content area is light gray and features the title 'Login' in a large font. Below the title, there is a link 'Don't have an account? [Sign up!](#)'. The login form consists of two input fields: 'Email address' with a placeholder 'Enter email' and 'Password' with a placeholder 'Password'. A blue 'Submit' button is positioned below the password field. The footer is dark and contains the copyright notice '© 2026 StockUp'.

Figure 5: Login Page



The screenshot shows the 'Sign Up' page of the 'StockUp' application. The header is dark with the 'StockUp' logo on the left and navigation links 'Home', 'Leaderboard', 'Invest', and a 'Login' button on the right. The main content area is light gray and features the title 'Sign Up' in a large font. Below the title, there is a link 'Already have an account? [Log in!](#)'. The sign-up form consists of five input fields: 'Email address' with a placeholder 'Email', 'Username' with a placeholder 'Username', 'Password' with a placeholder 'Password', and 'Confirm Password' with a placeholder 'Confirm Password'. A blue 'Submit' button is positioned below the 'Confirm Password' field. The footer is dark and contains the copyright notice '© 2026 StockUp'.

Figure 6: Sign Up Page

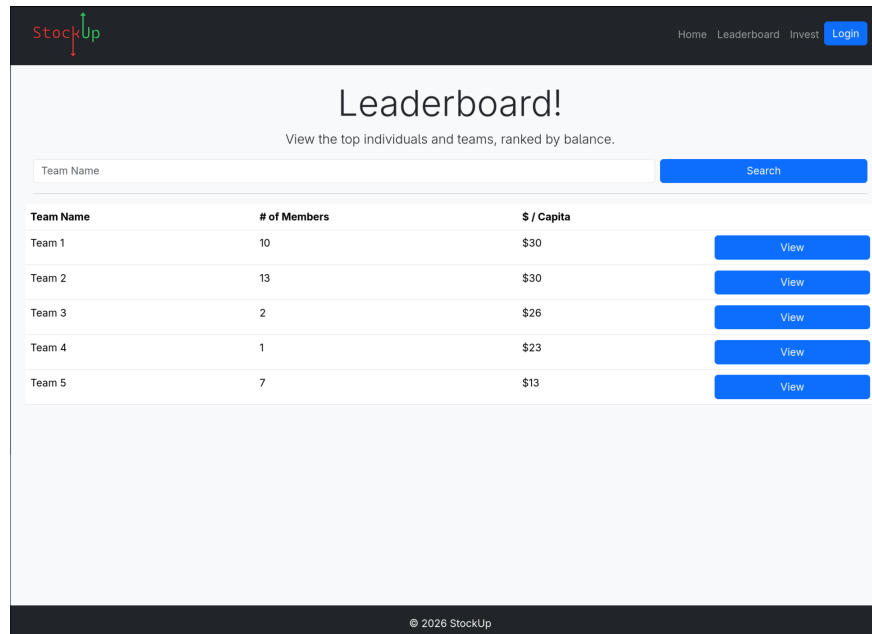


Figure 7: Leaderboards Page

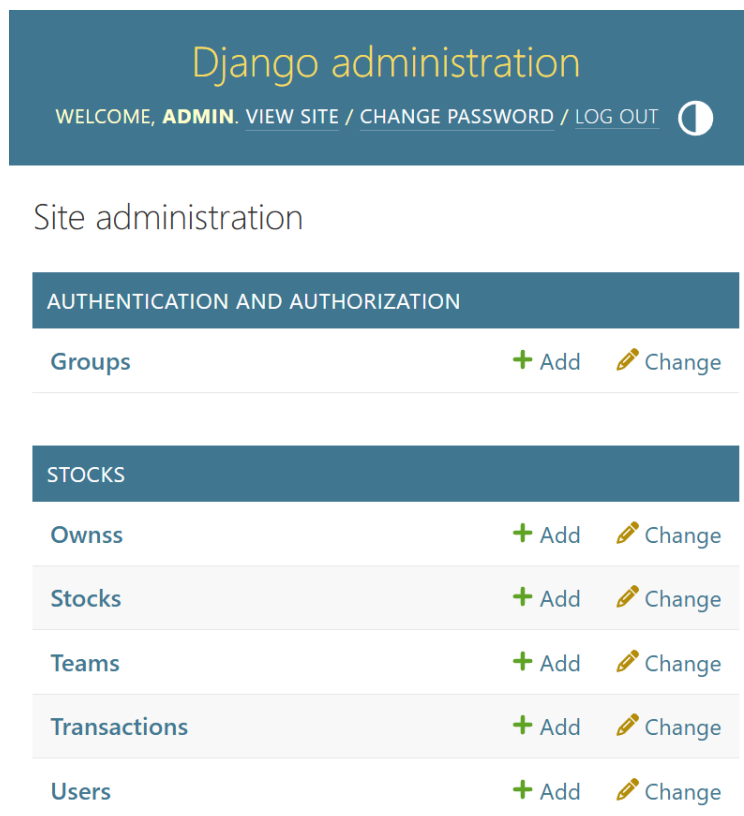


Figure 8: Database Administration Page

5 Timeline of Deliverables

- **April 15, 2026:** All parts of the project must work in isolation of each other. The website should be fully functional, the backend and API should be fully functional, and the database should be fully functional. At this point, the project should be done with Phase III.
- **May 1, 2026:** The project should be fully functional at this point. The required queries should be in place, and all aspects of the project should properly interact with minimal bugs. Phase IV should be complete at this point.
- **May 15, 2026:** The project must be complete by this point.